

## CS 170 Lab 2: Tracing Method Calls

1. Identify the parts of the following method:

```
public static double average(int x, int y) {  
    return (x + y) / 2.0;  
}
```

- a. What is the name of the method?

Average

- b. What is the return type?

Double

- c. What are the parameter types and names?

Int x and Int y (type- Int for both, names- x and y)

- d. What is the access modifier?

Public

- e. What is the body of the method?

return(x+y)/2.0;

2. Whenever a method is called,  
(1) the argument values passed are copied into the parameter variables,  
(2) program execution jumps to the method body,  
(3) when a return statement is reached, execution returns to the method call and replaces the method call with the returned value.

Write a line of code that is a method call to `average`, as defined above. Store the returned value in a variable of the appropriate type.

```
double avg1 = average(2, 3);
```

3. Consider this code:

```
1 public class MethodTracingLab {  
2     public static void main(String[] args) {  
3         int a = 3;  
4         int b = 4;  
5         int result = multiplyAndAdd(a, b);  
6         System.out.println("Final result: " + result);  
7     }  
  
8     public static int multiplyAndAdd(int x, int y) {  
9         int product = multiply(x, y);
```

```

10         int sum = add(product, y);
11         return sum;
12     }

13     public static int multiply(int n1, int n2) {
14         return n1 * n2;
15     }

16     public static int add(int n1, int n2) {
17         return n1 + n2;
18     }
19 }

```

Fill in the blanks in table below step by step to trace the program execution.  
There are 4 blanks indicated by ???.

| Step | Line Number                                                 | Method Call (and parameters)                          | Return Value              | Variable(s) Updated                       |
|------|-------------------------------------------------------------|-------------------------------------------------------|---------------------------|-------------------------------------------|
| 1    | 2                                                           | main                                                  | --                        | --                                        |
| 2    | 3                                                           | --                                                    | --                        | a = 3                                     |
| 3    | 4                                                           | --                                                    | --                        | b = 4                                     |
| 4    | 5 → 8                                                       | multiplyAndAdd(a,b)<br>x=3, y=4                       | --                        | --                                        |
| 5    | 9 → 13                                                      | multiply(x,y)<br>n1=3, n2=??? 4<br>4                  | --                        | --                                        |
| 6    | 14 → 9                                                      | --                                                    | 12 (3×4)                  | --                                        |
| 7    | 9                                                           | --                                                    | --                        | product=??? 12<br>product = multiply(3,4) |
| 8    | 10 → 16                                                     | ??? add(product, y)<br>product=12, y=4<br>n1=12, n2=4 | --                        | --                                        |
| 9    | 17 → ??? 10<br>17-body of add()<br>10-where add() is called | --<br>returned the value of add() to sum              | 16<br>add(12, 4)<br>→ sum | --                                        |
| 10   | 10                                                          | --                                                    | --                        | sum = 16                                  |
| 11   | 11 → 5                                                      | --                                                    | 16                        | --                                        |
| 12   | 5                                                           | --                                                    | --                        | result = 16                               |
| 13   | 6                                                           | System.out.println                                    | --                        | --                                        |

What is the program output (what is printed)?

Final result: 16

4. Consider this code:

```

1 public class MethodTracingPractice {
2     public static void main(String[] args) {
3         int x = 5;
4         int y = 2;
5         int result = squareDifference(x, y);
6         System.out.println("Answer: " + result);
7     }

8     public static int squareDifference(int a, int b){
9         int sq1 = square(a);
10        int sq2 = square(b);
11        return subtract(sq1, sq2);
12    }

13    public static int square(int n){
14        return n * n;
15    }

16    public static int subtract(int n1, int n2){
17        return n1 - n2;
18    }
19 }

```

Handwritten annotations and flow control:

- Line 3:  $x = 5$
- Line 4:  $y = 2$
- Line 5:  $5^2 = 25$  (above `square(a)`),  $2^2 = 4$  (above `square(b)`)
- Line 6:  $25 - 4 = 21$  (above `result`)
- Line 9:  $5^2 = 25$  (above `sq1`)
- Line 10:  $2^2 = 4$  (above `sq2`)
- Line 11:  $25 - 4 = 21$  (above `return`)
- Line 13:  $5^2 = 25$  (above `square`)
- Line 14:  $2^2 = 4$  (above `square`)
- Line 16:  $25 - 4 = 21$  (above `subtract`)
- Line 17:  $25 - 4 = 21$  (above `return`)

Flow control labels: A, B, C, D, E, F

Similar to the previous tracing table, complete this table to trace the above code.

| Step | Line Number | Method Call (and parameters)           | Return Value | Variable(s) Updated |
|------|-------------|----------------------------------------|--------------|---------------------|
| 1    | 2           | main                                   | --           | --                  |
| 2    | 3           |                                        |              | x = 5               |
| 3    | 4           |                                        |              | y = 2               |
| 4    | 5 → 8       | squareDifference(a, b)<br>x = 5, y = 2 |              |                     |
| 5    | 9 → 13      | square(a)<br>x = 5 $a = 5$             |              |                     |

|    |         |                                         |    |             |
|----|---------|-----------------------------------------|----|-------------|
| 6  | 14 → 9  |                                         | 25 |             |
| 7  | 9       |                                         |    | sq1 = 25    |
| 8  | 10 → 13 | square(b)<br>y = 2 <i>b = 2</i>         |    |             |
| 9  | 14 → 10 |                                         | 4  |             |
| 10 | 10      |                                         |    | sq2 = 4     |
| 11 | 11 → 16 | subtract(sq1, sq2)<br>sq1 = 25, sq2 = 4 |    |             |
| 12 | 17 → 11 |                                         | 21 |             |
| 13 | 11 → 5  |                                         | 21 |             |
| 14 | 5       |                                         |    | result = 21 |
| 15 | 6       | System.out.println                      |    |             |

What is the program output (what is printed)?

Answer: 21